Project Report: Fake News Truthfulness Classification (LIAR2)

1. Objective

Build and compare three machine learning models to classify political statements on a 6-level truthfulness scale, using the LIAR2 dataset from PolitiFact fact-checks.

2. Dataset

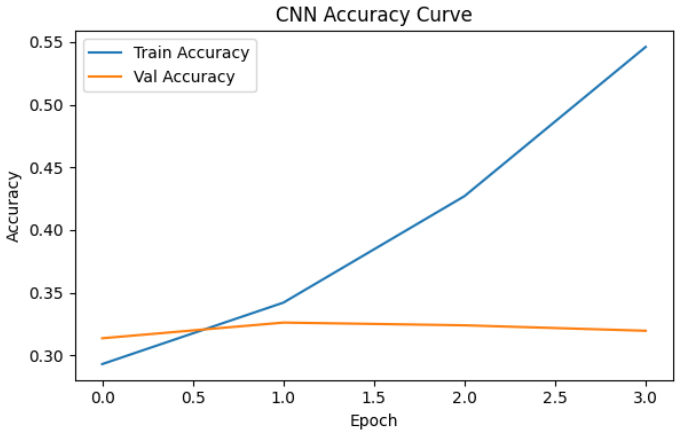
Source: LIAR2 (ucsbnlp/liar)
Training set: 18,369 statements
Test set: 2,296 statements
Classes are imbalanced - False (5,284) most common, True (2,068) least common.
Average statement length: 17.7 words. No missing values.

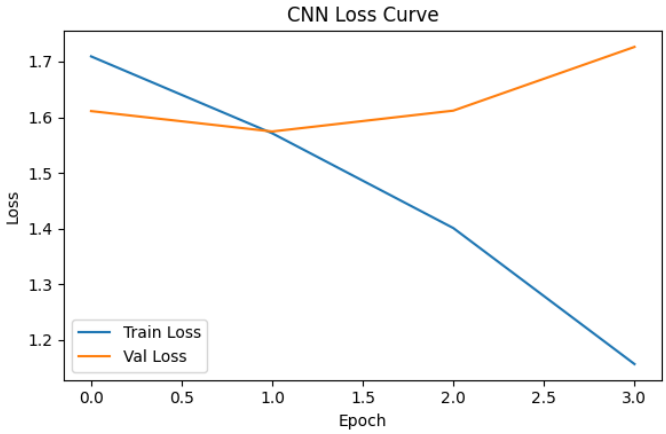
3. Methodology

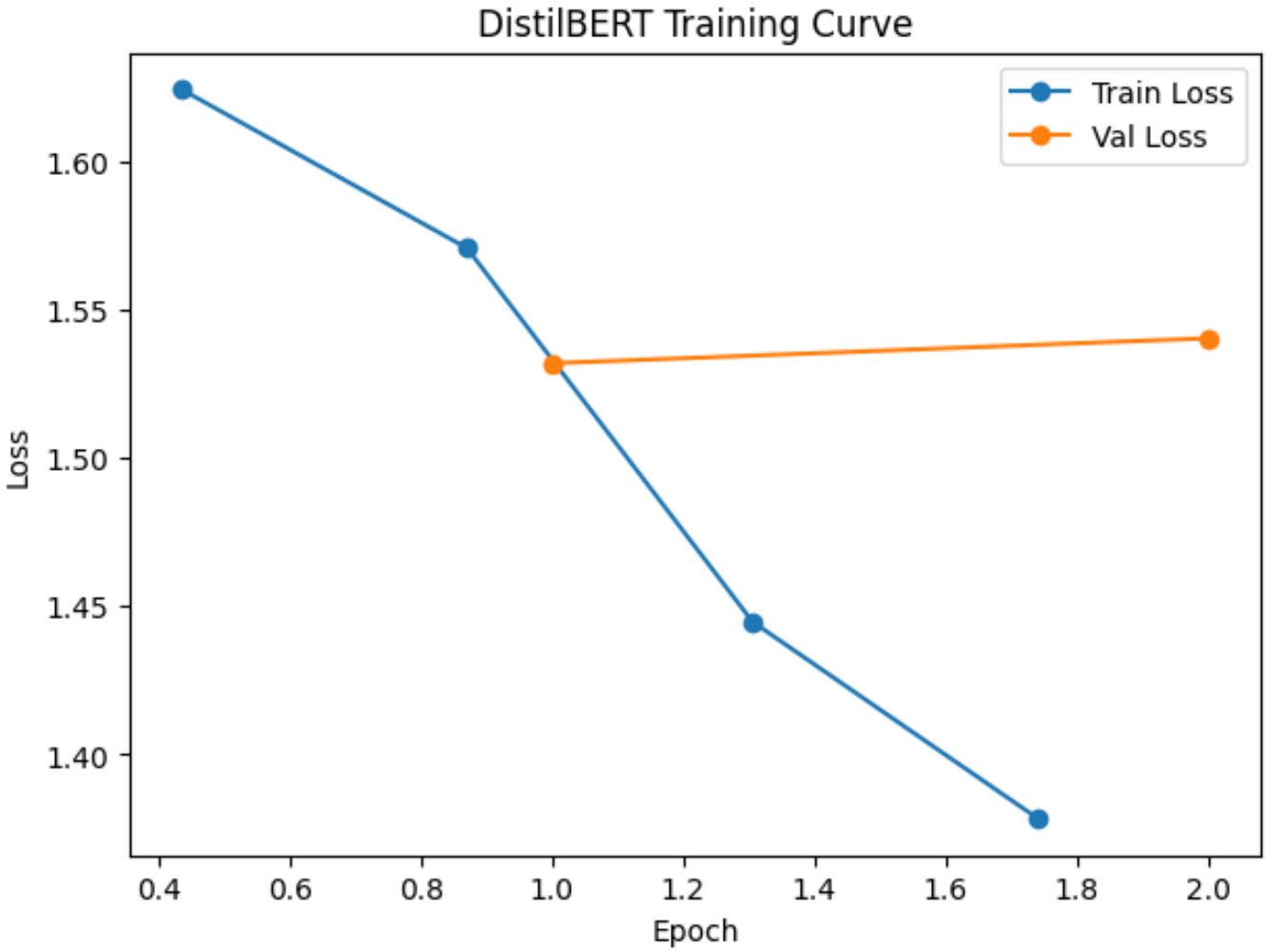
3.1 Logistic Regression - TF-IDF (5,000 features), sklearn LogisticRegression.

3.2 CNN - Embedding + Conv1D + GlobalMaxPooling, early stopping from the start (stopped epoch 4, restored epoch 2).

3.3 DistilBERT - Fine-tuned distilbert-base-uncased, 2 epochs, full 18,369-row dataset, ~3.5 hours on CPU. Mild overfitting observed (val_loss rose slightly).



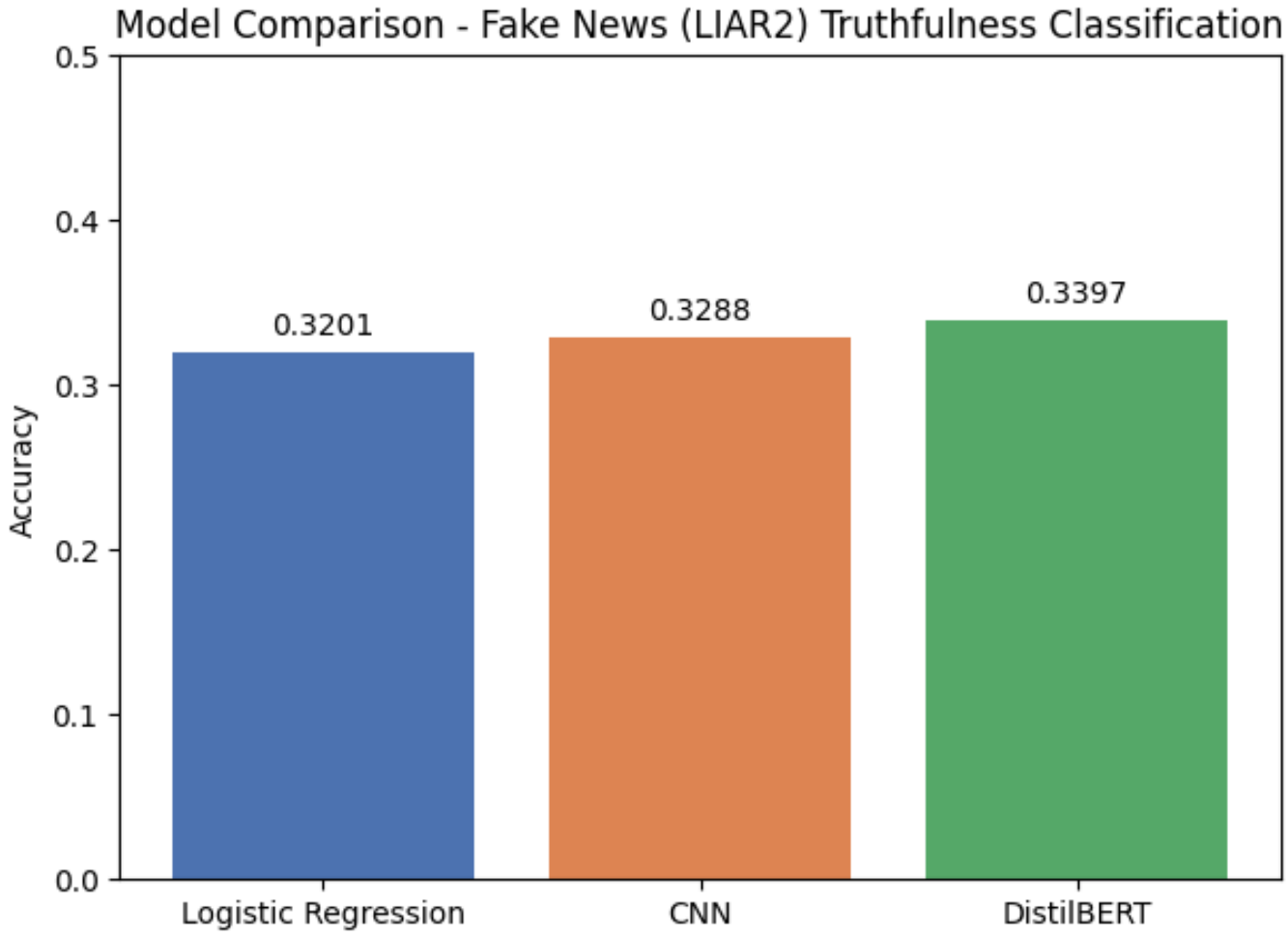




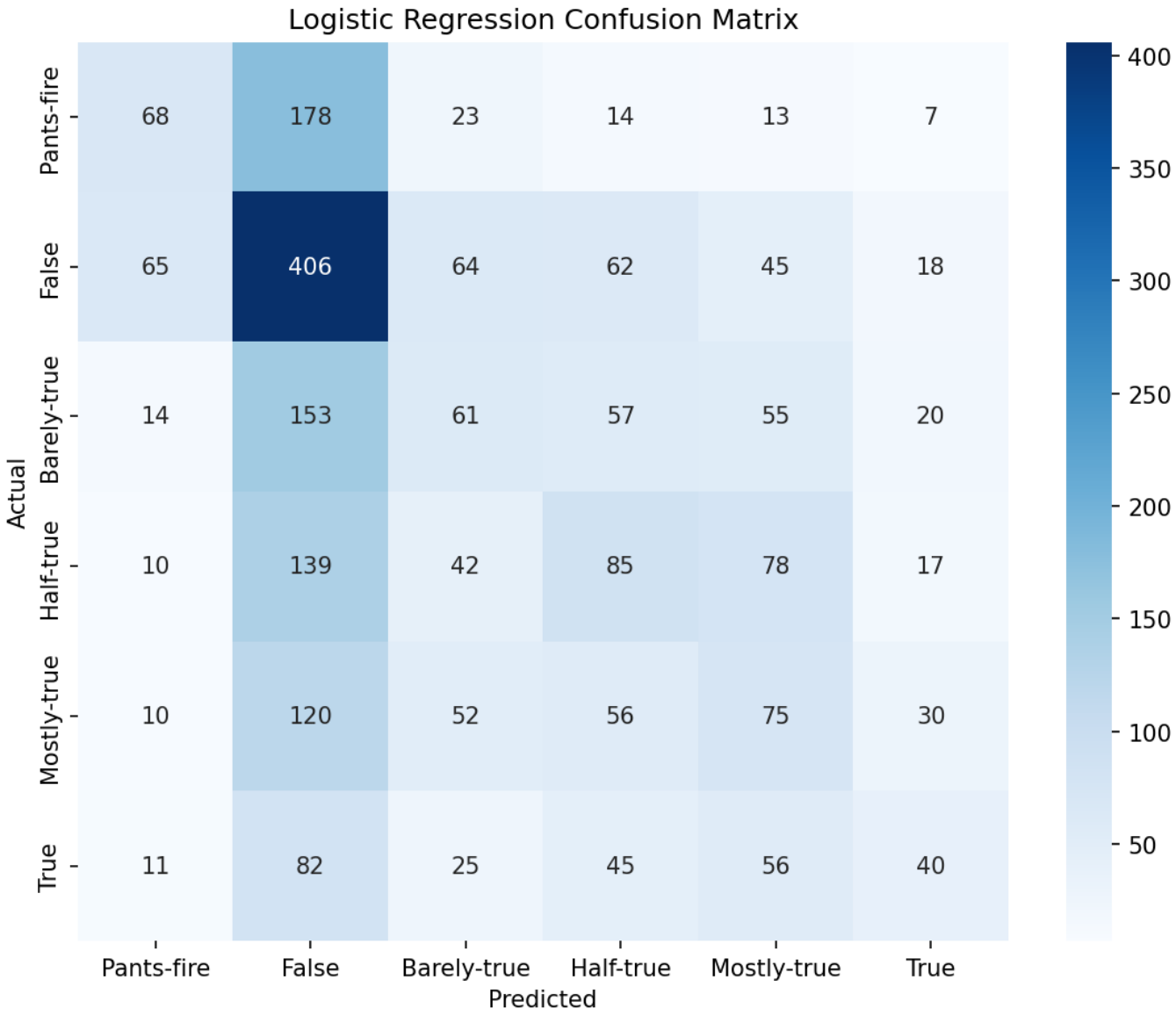
4. Results

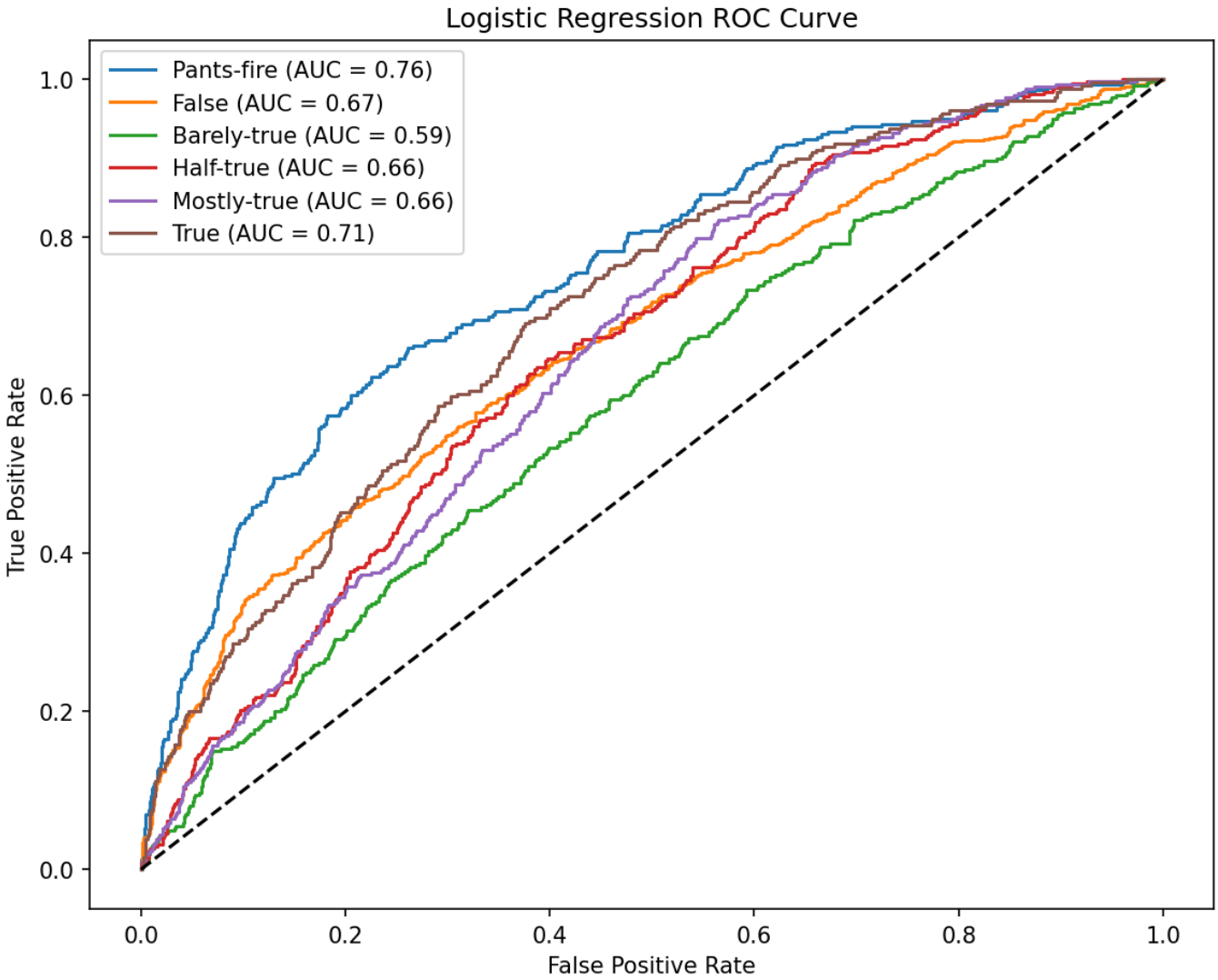
Logistic Regression: 32.01%
CNN: 32.88%
DistilBERT: 33.97%

For context, random guessing on 6 classes scores ~16.7%. Published statement-only LIAR benchmarks typically report 20-27% accuracy.

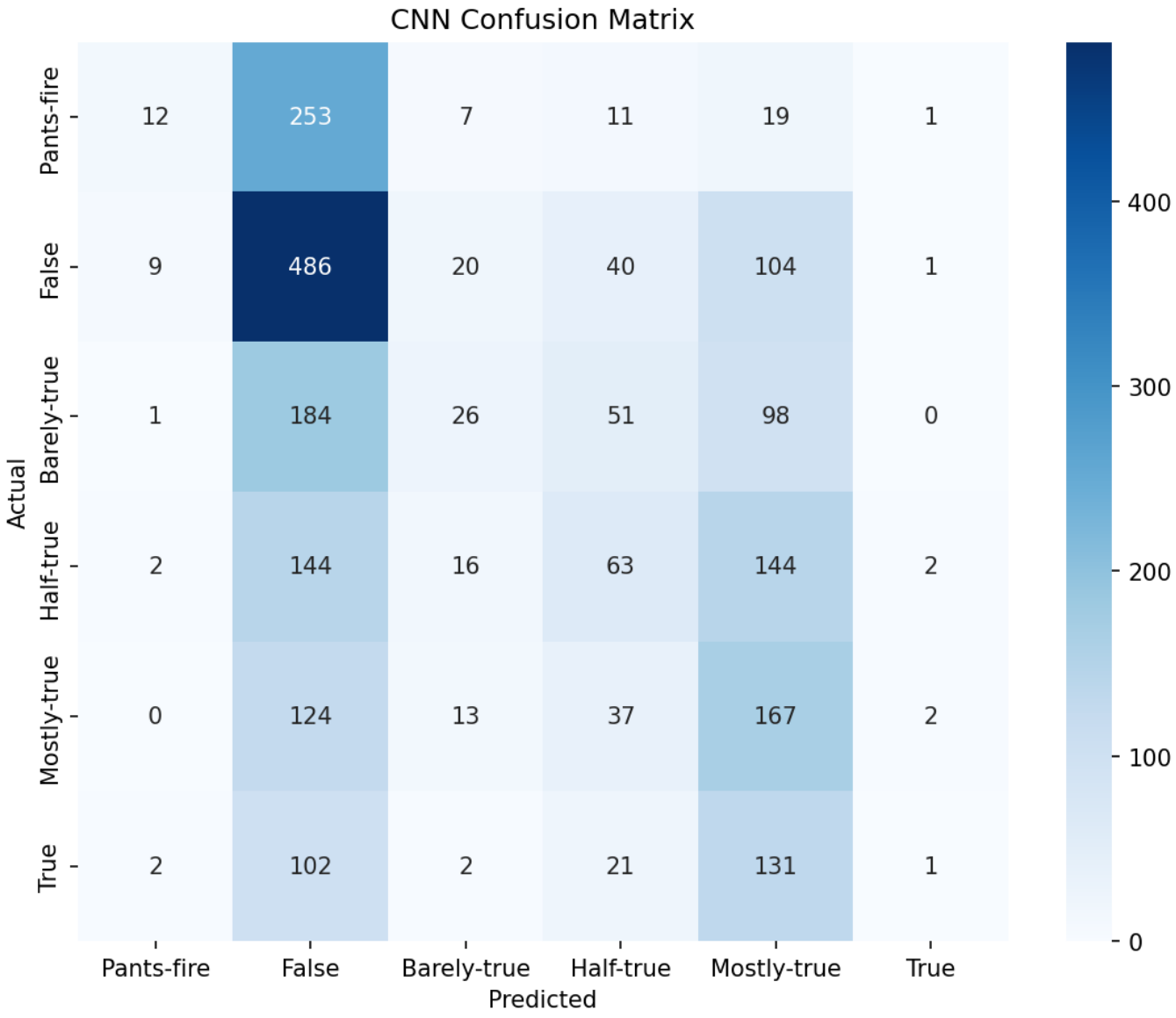


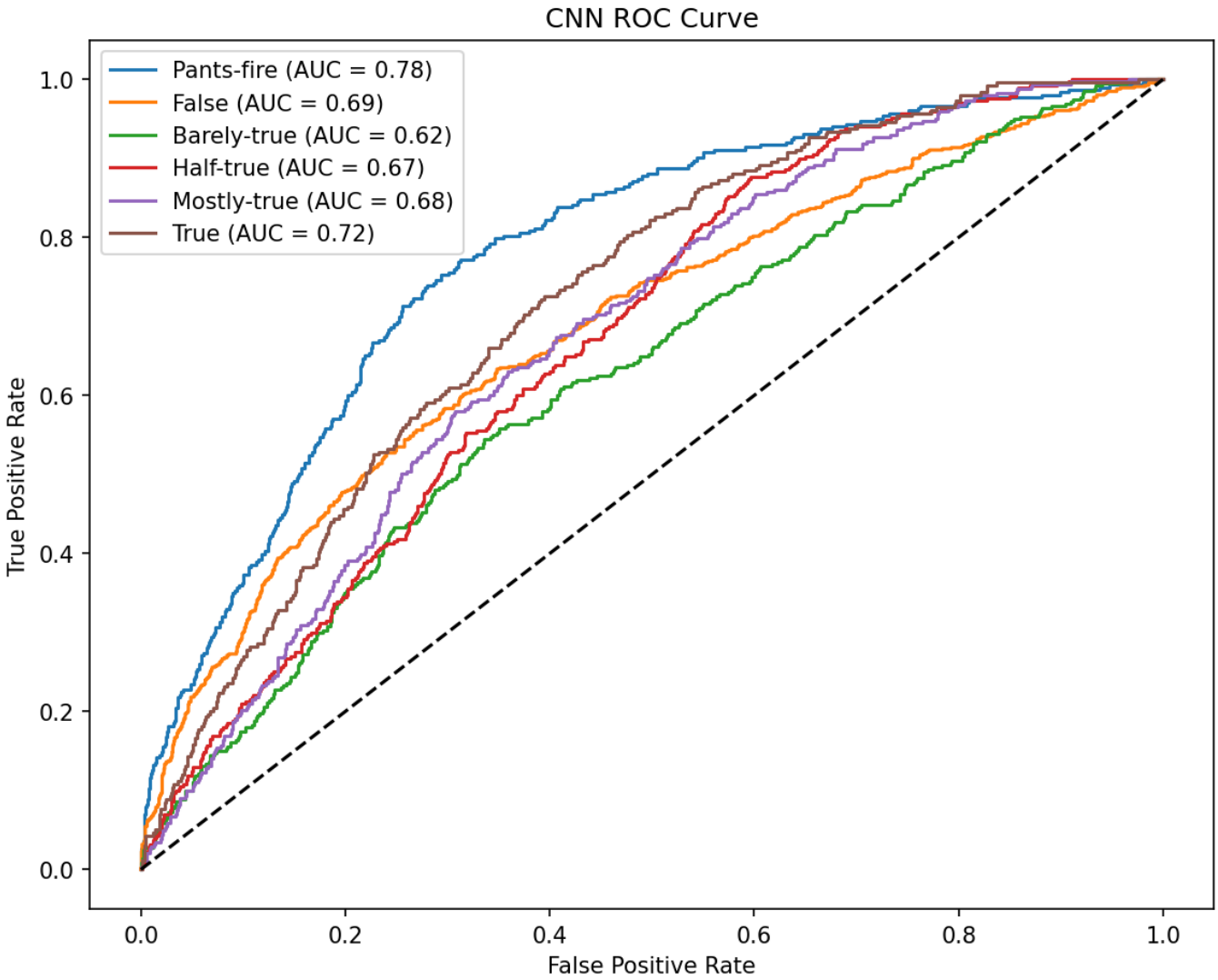
4.1 Logistic Regression


Logistic Regression Confusion Matrix
Actual
Pants-fire
False
Barely-true
Half-true
Mostly-true
True
68 178 23 14 13 7
65 406 64 62 45 18
14 153 61 57 55 20
10 139 42 85 78 17
10 120 52 56 75 30
11 82 25 45 56 40
Pants-fire
False
Barely-true
Half-true
Mostly-true
True
Predicted
400
350
300
250
200
150
100
50


Logistic Regression ROC Curve
Pants-fire (AUC = 0.76)
False (AUC = 0.67)
Barely-true (AUC = 0.59)
Half-true (AUC = 0.66)
Mostly-true (AUC = 0.66)
True (AUC = 0.71)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

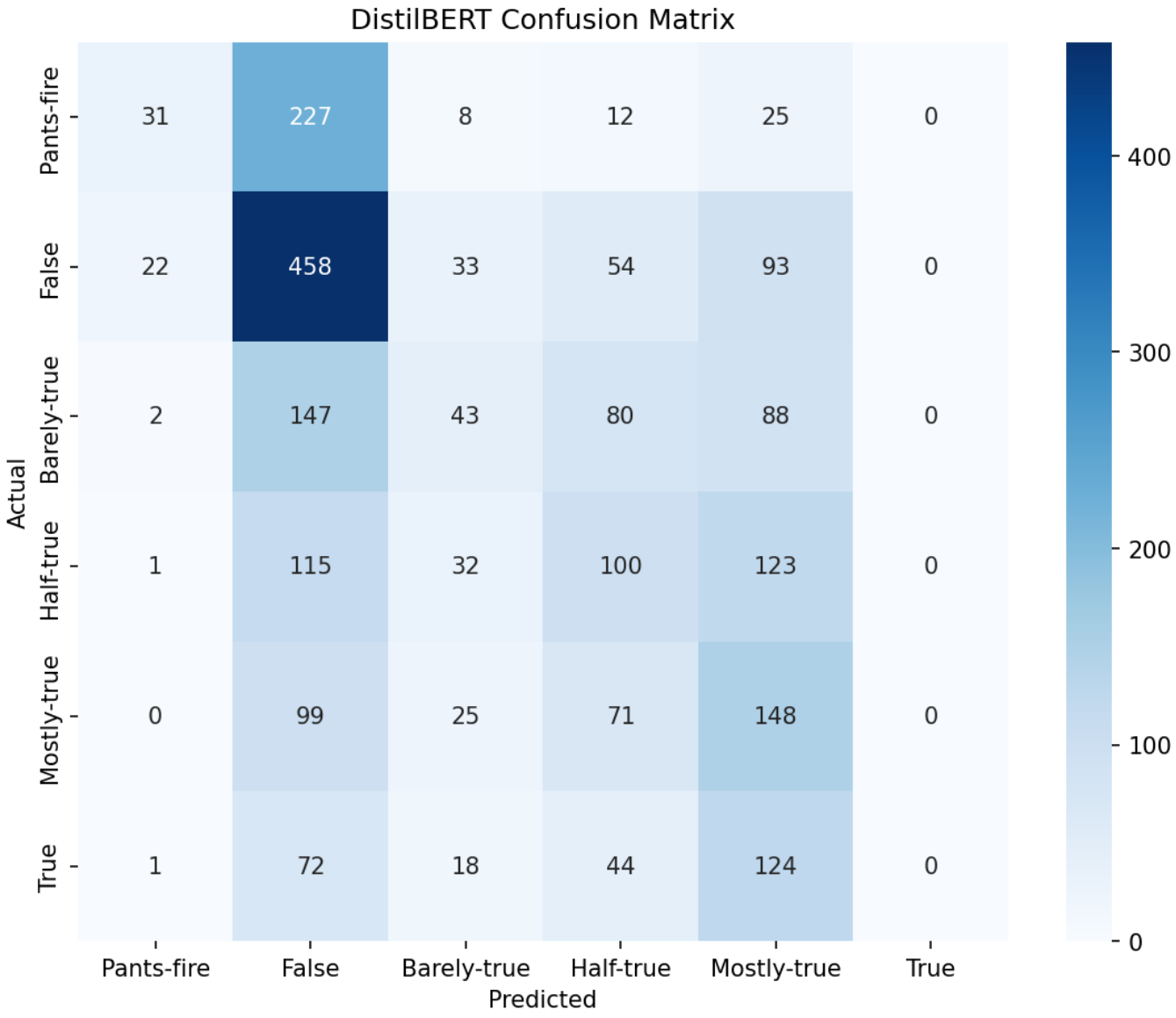
4.2 CNN

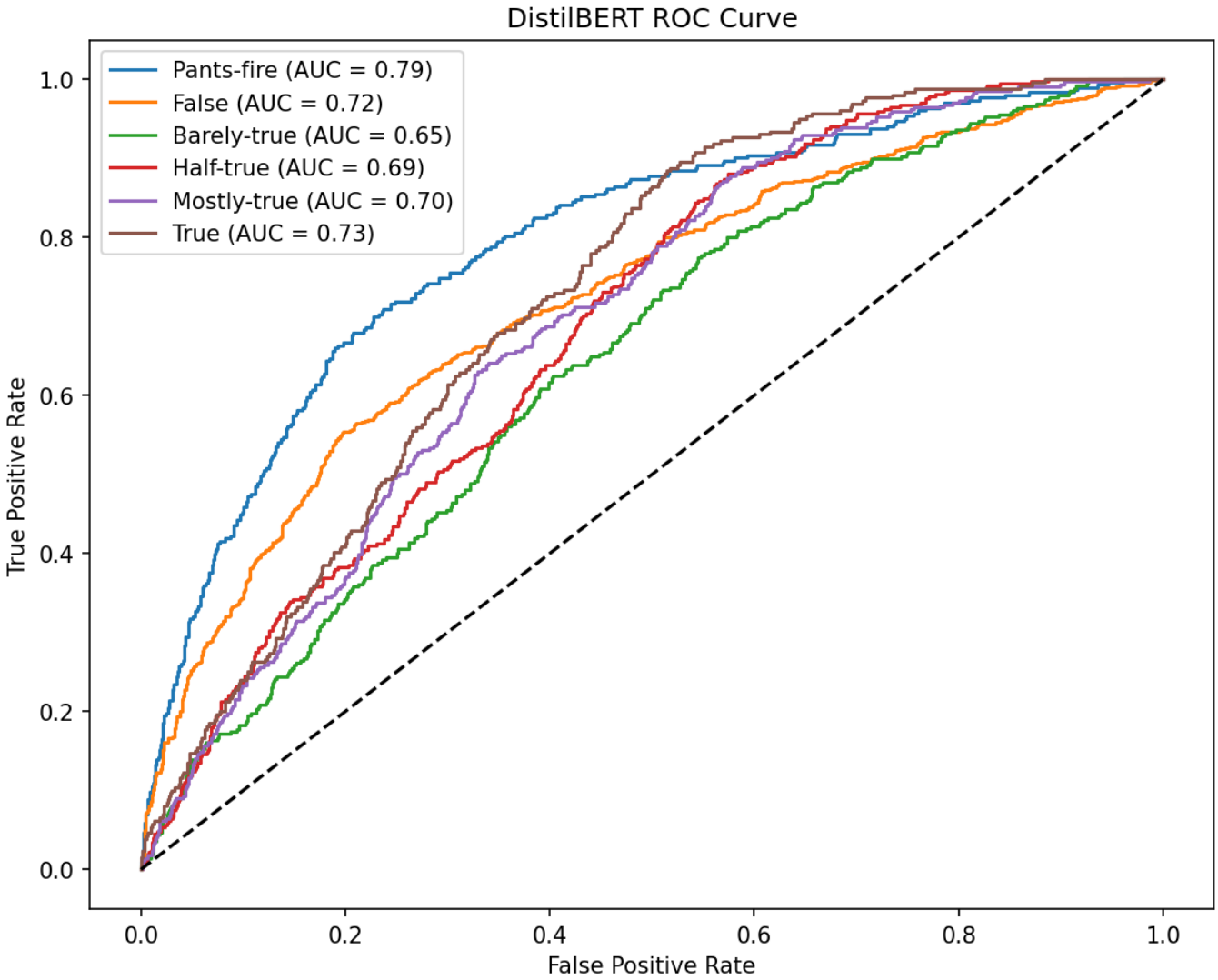

CNN Confusion Matrix
Actual
Pants-fire
False
Barely-true
Half-true
Mostly-true
True
12 253 7 11 19 1
9 486 20 40 104 1
1 184 26 51 98 0
2 144 16 63 144 2
0 124 13 37 167 2
2 102 2 21 131 1
Pants-fire
False
Barely-true
Half-true
Mostly-true
True
Predicted
400
300
200
100
0



4.3 DistilBERT

Notably, DistilBERT never correctly predicted the 'True' class (0.00 precision/recall), despite the highest overall accuracy - it leaned heavily on the majority 'False' class instead.


DistilBERT Confusion Matrix
Actual
Predicted
Pants-fire
False
Barely-true
Half-true
Mostly-true
True
31 227 8 12 25 0
22 458 33 54 93 0
2 147 43 80 88 0
1 115 32 100 123 0
0 99 25 71 148 0
1 72 18 44 124 0
400
300
200
100
0



5. Discussion

This is a genuinely difficult task: distinguishing six truthfulness levels from short text alone requires real-world fact verification the models cannot access. All three models performed similarly, suggesting the bottleneck is information availability, not model capacity. Class imbalance likely contributed to DistilBERT's failure on the 'True' class, underscoring the importance of per-class metrics over overall accuracy alone.

6. Conclusion

DistilBERT achieved the highest accuracy but failed completely on the 'True' class, making it an imperfect choice without further adjustment. Logistic Regression and CNN performed comparably with more balanced per-class behavior.

7. Future Work

- Apply class weighting to address the 'True' class collapse
- Incorporate speaker history, justification, and subject fields
- Try a simplified 3-class version of the task
- Deploy the best, rebalanced model as a web app